

CMP-09 - Memory System

Maintain contextual associations and governed retrieval of retained experience without confusing memory, knowledge, cache, or authoritative state.

Authority and state ownership

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 12. Authoritative state: Memory Graph, memory associations, retention and access metadata, Conversation Objects where assigned. Explicitly non-owned: Knowledge Graph, Evidence Objects, Experience Ledger custody, identity or authority.

Critical invariant

Memory is provenance-linked, purpose-bound, and never silently promoted to knowledge or evidence.

Logical interfaces

CMP-09-IF-01 / Command / Store Memory Association

Command `Store Memory Association` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-02 / Command / Revise Association

Command `Revise Association` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-03 / Command / Forget Memory

Command `Forget Memory` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-04 / Command / Reconcile Conversation

Command `Reconcile Conversation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-05 / Query / Recall Memory

Query `Recall Memory` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-06 / Query / Trace Memory Source

Query `Trace Memory Source` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-07 / Query / Get Retention Status

Query `Get Retention Status` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-08 / Event / Memory Associated

Event `Memory Associated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-09 / Event / Memory Forgotten

Event `Memory Forgotten` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-09-IF-10 / Event / Conversation Reconciled

Event `Conversation Reconciled` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Candidate > Active > Restricted > Expired or Deleted. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: cross-principal leakage, retention violation, false association, conversation split, deletion failure. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-09-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-09-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-09-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-09-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-09-TST-005 - Critical invariant. Memory is provenance-linked, purpose-bound, and never silently promoted to knowledge or evidence. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-09-TST-006 - Failure containment. Inject cross-principal leakage, retention violation, false association, conversation split, deletion failure. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-09-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-09-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-09-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-09-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.